

MICROCANONICAL EXCESS-PRESSURE SEMIGROUPS AND KINETIC THETA-GENERATORS

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ABSTRACT. Microscopic states, Koopman observables, pushed-forward laws, initial-law endpoint values, and the limiting action semigroup are kept distinct. A resolvent image of smooth reflected observables is a genuine generator core, and the full hierarchy corrector is solved in an exponential generating sequence space. The dynamic action is compared by primal verification along finite-entropy controls, avoiding uncontrolled exponential doubled jets.

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1. ROUND-TWELVE POSITIVE CLOSURE: TYPED ENDPOINT VALUES, RESOLVENT CORES, AND PRIMAL COMPARISON

Five levels remain distinct: microscopic states, Koopman observables, push-forward laws, endpoint values depending on an initial law, and the limiting kinetic action semigroup. Uniqueness of the limiting Hamilton–Jacobi equation is proved from the primal entropy action, not from a doubled penalty whose exponential jets escape every controlled source ball.

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1.1. The exact finite-volume objects. Let Φ_t^ε be the hard-sphere flow, $U_t^\varepsilon F = F \circ \Phi_t^\varepsilon$ its Koopman group, and $(U_t^\varepsilon)_\# \rho$ the push-forward of a law. For a terminal microscopic observable G define

$$V_{s,t}^\varepsilon(G; \rho) = \frac{1}{\mu_\varepsilon} \log \int e^{\mu_\varepsilon U_{t-s}^\varepsilon G(x)} d\rho(x).$$

Proposition 1.1 (Typed endpoint tower). *For $r < s < t$,*

$$V_{r,t}^\varepsilon(G; \rho) = \frac{1}{\mu_\varepsilon} \log \int \exp\{\mu_\varepsilon V_{s,t}^\varepsilon(G; \delta_{\Phi_{s-r}^\varepsilon x})\} d\rho(x).$$

The inner function is a microscopic observable; no finite-volume Markov closure on the one-particle density is asserted.

Proof. Substitute the definition and use $U_{t-r} = U_{s-r} U_{t-s}$. The types of the inner Dirac endpoint value and the outer initial-law integral are exactly those displayed. $\square$

1.2. A genuine observable graph core. Let A_ε be the closed Koopman generator on the weighted uniformly continuous observable space with specular boundary matching. Let $\mathcal{D}_0$ be smooth finite-coordinate functions satisfying that matching, and define

$$\mathcal{C}_\varepsilon = R_{\lambda,\varepsilon} \mathcal{D}_0, \quad R_{\lambda,\varepsilon} = (\lambda - A_\varepsilon)^{-1}.$$

Theorem 1.2 (Resolvent graph core). *$\mathcal{D}_0$ is dense in the underlying observable Banach space and $\mathcal{C}_\varepsilon$ is a core for A_ε . More precisely, for $F \in D(A_\varepsilon)$ and $G = (\lambda - A_\varepsilon)F$, any $G_n \in \mathcal{D}_0$ with $G_n \rightarrow G$ gives $R_{\lambda,\varepsilon} G_n \rightarrow F$ in graph norm.*

Proof. Unfold specular trajectories across each collision face. On every finite chronological chart the flow is translation in unfolded coordinates, so mollification and velocity truncation approximate bounded uniformly continuous observables while preserving the reflected matching after folding back. Grazing and multiple-collision sets are bypassed in the base norm, not claimed to have zero graph capacity. This proves density of $\mathcal{D}_0$. The resolvent identity then gives $AR_\lambda G_n = \lambda R_\lambda G_n - G_n$, which is graph convergence. $\square$

1.3. The full hierarchy corrector in one sequence space. Use the exponential generating hierarchy norm

$$\|G\|_\alpha = \sum_{j \geq 0} \frac{e^{\alpha j}}{j!} \|g_j\|_{j,\beta}.$$

Let $\mathbb{A}_\varepsilon$ be the full triangular BBGKY/ledger generator on its closed graph domain.

Theorem 1.3 (Normally summable graph corrector). *For every smooth density cylinder F and $T \leq T_*$ there is a unique sequence-space perturbed test $F_\varepsilon^{\text{corr}}$ satisfying the backward graph equation*

$$(\partial_t + \mathbb{A}_\varepsilon) F_\varepsilon^{\text{corr}} = \mathbb{H}(f, DF, \psi) + r_\varepsilon, \quad F_\varepsilon^{\text{corr}}(T) = F,$$

with

$$\sup_{t \leq T} (\|F_\varepsilon^{\text{corr}} - F\|_\alpha + \|r_\varepsilon\|_{\alpha'}) \longrightarrow 0.$$

The factorial tree growth is cancelled explicitly by the $1/j!$ hierarchy weight and the remaining series is geometric for short time.

Proof. Write the Duhamel equation on the complete weighted sequence space. A connected order- j term is at most $j!C^jT^{j-1}$; multiplication by the sequence weight leaves $(Ce^{\alpha-\alpha'}T)^j$. Choose T_* so this is summable, including one generator image and time derivative. B2 removes the summed cyclic sector and identifies the tree limit. Closedness of $\mathbb{A}_\varepsilon$ passes the normally convergent series through the graph equation. $\square$

1.4. The dynamic action semigroup. Let $I_{s,t}^{\text{dyn}}$ be only the B2 collision entropy on $[s, t]$, with no initial preparation term. Define

$$S_{s,t}\Phi(f) = \sup_{(\rho,\Gamma):\rho_s=f} \{\Phi(\rho_t) - I_{s,t}^{\text{dyn}}(\rho, \Gamma)\}.$$

Theorem 1.4 (Action compactness and dynamic programming). *On every moment sublevel, $I_{s,t}^{\text{dyn}}$ has compact sublevels, is lower semicontinuous, and is additive under concatenation. Hence $S_{r,t} = S_{r,s}S_{s,t}$ and S is a sup-norm contraction. Primitive microcanonical preparation is charged once through the B1 initial rate.*

Proof. B2 exponential moments and the Green balance give weighted tightness and endpoint compactness. Entropy lower semicontinuity and closed balance give closed sublevels. Concatenating balanced pairs adds the entropy integral; splitting an admissible pair gives the reverse dynamic-programming inequality. The initial rate never appears in a transition interval. $\square$

1.5. Comparison from the primal control problem. A bounded upper semicontinuous u is a subsolution when every smooth bounded cylinder test gives the Hamiltonian inequality. Supersolutions are analogous.

Theorem 1.5 (Primal verification comparison). *Let u be a bounded subsolution and v a bounded supersolution with the same terminal data and the common moment-containment condition. Then $u \leq v$. The proof never evaluates the Hamiltonian on an uncontrolled doubled jet.*

Proof. Fix a smooth positive control pair (ρ, Γ) with bounded collision ratio q . Fenchel's inequality applied to the subsolution test gives, after partitioning time and using the chain rule along the weak balance,

$$u(s, \rho_s) \leq u(t, \rho_t) + I_{s,t}^{\text{dyn}}(\rho, \Gamma) + o(1).$$

For a supersolution, use the regular tilted recovery supplied by B2 and the reverse local inequality to obtain

$$v(s, f) \geq \sup_{(\rho,\Gamma):\rho_s=f} \{v(t, \rho_t) - I_{s,t}^{\text{dyn}}(\rho, \Gamma)\}.$$

Approximate finite-action controls by the B2 positive regularization; entropy uniform integrability removes the bounded- q restriction. Iterating the two inequalities to the terminal time gives $u \leq S\Phi \leq v$. No exponential continuity estimate at a diverging cotangent is used. $\square$

Theorem 1.6 (Microscopic nonlinear-semigroup convergence). *The finite-volume endpoint values converge locally uniformly on regular moment sets to $S_{s,t}\Phi$. The limit is the unique viscosity/semigroup solution of*

$$\partial_s U + \mathbb{H}(f, DU, \psi) = 0$$

in the preceding class. Under B3 central scaling it converges to the risk-sensitive Gaussian semigroup with quadratic term $\frac{\theta}{2} DU^T A DU$.

Proof. Theorem 1.3 gives upper and lower nonlinear-generator limits on the resolvent core. B2 compact containment gives half-relaxed limits, and Theorem 1.5 identifies them with the action value. The exact endpoint tower and local uniform convergence give the semigroup law. B3 Gaussian convergence and a uniform complex moment neighborhood give the final exponential-transform limit. $\square$

Theorem 1.7 (Round-twelve B4 closure). *The exact microscopic objects are correctly typed, the Koopman generator has a resolvent graph core, the full hierarchy corrector is normally summable, and the kinetic action semigroup is the unique limit by primal comparison.*

Proof. Combine Proposition 1.1 and Theorems 1.2, 1.3, 1.4, 1.5, and 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-twelve dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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